WORKSHEET

Chapter 9: Numerical integration

1 Calculate an approximation for the definite integral of $f(x) = \frac{1}{4}x^3$ from $x = -0.5$ to $x = 3.5$ with strips 1 unit wide, using the midpoint rule.

2 Calculate an approximation for $\int_0^2 e^{x^2}\,dx$ with strips 0.5 units wide, using the midpoint rule.

3 Calculate an approximation for the area under the curve $y = \frac{1}{x}$ from $x = 1$ to $x = 2$ with strips 0.2 units wide, using:

a the midpoint rule

b the trapezium rule

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4 Calculate an approximation for $\int_0^1 \sqrt{1+x^3}\,dx$ with strips 0.1 units wide, using:

a the midpoint rule

b the trapezium rule

5 Calculate an approximation for $\int_0^1 e^{x^2}\,dx$ with strips 0.1 units wide, using:

a the midpoint rule

b the trapezium rule

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6 Calculate an approximation for $\int_1^6 (x^4 - 3x^2)dx$ with strips 0.5 units wide, using:

a the trapezium rule

b Simpson's rule

7 Calculate an approximation for $\int_{-1}^{1} \frac{dx}{x+2}$ with strips 0.2 units wide, using:

a the trapezium rule

b Simpson's rule

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8 Calculate an approximation for $\int_0^3 2^x dx$ with strips 0.25 units wide, using:

a the trapezium rule

b Simpson's rule

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Answers

1 9

2 14.486

3 **a** 0.6919

b 0.6956

4 **a** 1.1110

b 1.1123

5 **a** 1.4604

b 1.4672

6 **a** 1357.28

b 1340.04

7 **a** 1.1016

b 1.0987

8 **a** 10.1241

b 10.0989

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